

Way to go! You really deserve that high score!

Your time: 5 mins



Correct

24

Incorrect

0

Next steps

Take a new test



Try another test to boost your confidence.

Review with Flashcards



Study each term as a flashcard to improve recall.

Your answers

1 of 24

Term 

What are the **2 Diencephalon structures** we need to know?

You're doing great!

1. Corticospinal tracts

2. Corticobulbar tracts

1. Tectum

2. Tegmentum

3. Cerebral Aqueduct

Review flashcards

New test

dorsal side of the midbrain

1. Thalamus
2. Hypothalamus

Don't know?

2 of 24

Term 🔊

Midbrain Structures:

What is the function of the Tectum?

Nice work!

emotional reactions to visual or auditory cues

cognitive processing of visual and auditory information

voluntary movements in response to sensory stimuli

reflex actions in connection with visual or auditory stimuli

Don't know?

Term 🔊

Midbrain Structures:

What does the Tegmentum consist of?

Awesome!

1. **Reticulospinal tract** (exerts some control over alertness and takes input from the tectum)
2. **The superior colliculi** (process visual information, aid the decussation of optic nerve fibers, and are involved in saccadic eye movements)
3. **The inferior colliculi** (process certain auditory information)

1. Red Nuclei
(Gait)
2. Medial Lemniscus
(somatosensation from the skin and joints)
3. Reticular Formation
(production of 5-HT, arousal, consciousness, motor coordination, & exhalation)

1. Connects the 3rd and 4th ventricles (the smallest ventricle in the ventricular system)
2. Responsible for continuing the circulation of cerebrospinal fluid

between the tectum and tegmentum and is surrounded by the **periaqueductal grey** (which has a role in analgesia, bonding, etc.)

Definition 🔊

1. **Reticulospinal tract** (exerts some control over alertness and takes input from the tectum)
2. **The superior colliculi** (process visual information, aid the decussation of optic nerve fibers, and are involved in saccadic eye movements)
3. **The inferior colliculi** (process certain auditory information)

Excellent!

Midbrain Structures:

What does the Tectum consist of?

Midbrain Structures:

What is the function of the Tectum?

Midbrain Structures:

What is the function of the Cerebral Aqueduct?

Midbrain Structures:

What does the Tegmentum consist of?

Don't know?

Term 🔊

Spinal Pathways:

What are the "descending extrapyramidal tracts?"

You've got this!

1. Dorsal column-medial lemniscal pathway
2. Anterolateral system

1. Vestibulospinal tracts
2. Reticulospinal tracts
3. Rubrospinal tracts
4. Tectospinal tracts

1. Endocrine control via the Pituitary Gland
2. Control of eye movement and hearing responses

1. Corticospinal tracts
2. Corticobulbar tracts

Don't know?

Term 🔊

Midbrain Functions:

What are the midbrain's *associated autonomic functions*?

language processing, balance
coordination, hunger regulation, &
heart rate control

cognitive reasoning, muscle growth,
pain perception, & blood pressure
regulation

motor-control, sleep/wake,
arousal/alertness, &
temperature-regulation

sensory processing, memory storage,
emotional regulation, & digestion
control

Don't know?

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Term 🔊

Midbrain Structures:

What is the function of the Cerebral Aqueduct?

Nice work!

1. contains the corticobulbar and corticospinal tracts
2. other tracts connecting the cortex to the pons
3. substantia nigra (the only part of the basal ganglia system outside of the forebrain)
 - Pars compacta and pars reticulata
 - Motor planning, learning, addiction, and other functions
 - High production of melanin, dopamine, and noradrenaline

between the tectum and tegmentum and is surrounded by the **periaqueductal grey** (which has a role in analgesia, bonding, etc.)

1. Connects the 3rd and 4th ventricles (the smallest ventricle in the ventricular system)
2. Responsible for continuing the circulation of cerebrospinal fluid

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(Gait)
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(somatosensation from the skin and joints)
3. Reticular Formation
(production of 5-HT, arousal, consciousness, motor coordination, & exhalation)

Don't know?

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Term 🔊

Diagram: "Mesencephalon" Cross-Section at the Level of the Superior Colliculus

Awesome!

1. Midbrain= above pons
2. Pons= between midbrain and medulla
3. Medulla= below pons



1. Nucleus of inferior colliculus
2. Brachium of inferior colliculus
3. Cerebral aqueduct
4. Lateral lemniscus
5. Medial longitudinal fasciculus (MLF)
6. Trochlear nucleus
7. Central tegmental tract
8. Trigeminal and spinal lemnisci
9. Reticular formation of the tegmentum
10. Decussation of superior cerebellar peduncles

11. Medial lemniscus
1. Superior colliculus
12. Pontine nuclei and pontocerebellar fibers
2. Brachium of superior colliculus
13. Corticospinal, corticopontine, and corticobulbar fibers (these collectively make up the Crus Cerebri or Cerebral Peduncle at this level)
4. Brachium of inferior colliculus (This part of the inferior colliculus's pathway is still visible here)
5. Central gray substance (Periaqueductal Gray)
6. Cerebral aqueduct
7. Edinger-Westphal nucleus (Accessory oculomotor nucleus)
8. Nucleus of oculomotor nerve (CN III nucleus)
9. Medial lemniscus
10. Central tegmental tract
11. Medial longitudinal fasciculus (MLF)
12. Red nucleus
13. Fibers of oculomotor nerve (CN III fibers)
14. Substantia nigra
15. Basis pedunculi or "Crus Cerebri" (which contains the Corticospinal, Corticopontine, and Corticobulbar fibers).

1. Superior Colliculus
2. Cerebral Aqueduct
3. Oculomotor Nucleus
4. Red Nucleus
5. Spinothalamic Tract and Medial Lemniscus (often grouped together in this view)
6. Substantia Nigra
7. Crus Cerebri (or Cerebral Peduncle)

[Don't know?](#)

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Definition

1. Superior colliculus
2. Brachium of superior colliculus
3. Medial geniculate nucleus
4. Brachium of inferior colliculus (This part of the inferior colliculus's pathway is still visible here)
5. Central gray substance (Periaqueductal Gray)
6. Cerebral aqueduct
7. Edinger-Westphal nucleus (Accessory oculomotor nucleus)
8. Nucleus of oculomotor nerve (CN III nucleus)
9. Medial lemniscus
10. Central tegmental tract
11. Medial longitudinal fasciculus (MLF)
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13. Fibers of oculomotor nerve (CN III fibers)
14. Substantia nigra
15. Basis pedunculi or "Crus Cerebri" (which contains the Corticospinal, Corticopontine, and Corticobulbar fibers).

Excellent!

Diagram: Rostral midbrain

Diagram: "Mesencephalon" Cross-Section at the Level of the Superior Colliculus

Midbrain Structures:

What does the Tectum consist of?

Diagram: Caudal midbrain

Don't know?

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Term 🔊

What are the **4 Midbrain structures** we need to know?

You've got this!

1. Tectum
2. Tegmentum
3. Cerebral Aqueduct
4. Crus Cerebri (Basis Pedunculi)

1. Vestibulospinal tracts
2. Reticulospinal tracts
3. Rubrospinal tracts
4. Tectospinal tracts

1. Dorsal column-medial lemniscal pathway
2. Anterolateral system

1. Thalamus
2. Hypothalamus

Don't know?

Term 🔊

Diencephalon Structures:

What is the function of the Hypothalamus?

You're doing great!

1. subthalamus

2. epithalamus

--serves as a relay center for sensory and motor impulses between the spinal cord, medulla, and cerebrum

--recognizes sensory impulses of heat, cold, pain, pressure, vision, taste, balance...NOT smell

1. Vestibulospinal tracts

2. Reticulospinal tracts

3. Rubrospinal tracts

4. Tectospinal tracts

1. Endocrine control via the Pituitary Gland

2. Control of eye movement and hearing responses

1. Dorsal column-medial lemniscal pathway

2. Anterolateral system

Don't know?

Definition 🔊

1. contains the corticobulbar and corticospinal tracts
2. other tracts connecting the cortex to the pons
3. substantia nigra (the only part of the basal ganglia system outside of the forebrain)
 - Pars compacta and pars reticulata
 - Motor planning, learning, addiction, and other functions
 - High production of melanin, dopamine, and noradrenaline

Nice work!

Midbrain Structures:

What does the Tectum consist of?

Midbrain Structures:

What does the "Crus Cerebri" (Basis Pedunculi) consist of? (and its function?)

Midbrain Structures:

What is the function of the Cerebral Aqueduct?

Midbrain Structures:

What does the Tegmentum consist of?

Don't know?

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Term 🔊

Midbrain Structures:

Where is the Cerebral Aqueduct located?

Awesome!

beside the cerebellum and is surrounded by cerebellar nuclei

below the tegmentum and links to the spinal cord

between the tectum and tegmentum and is surrounded by the periaqueductal grey (which has a role in analgesia, bonding, etc.)

above the tectum and connects to the cerebral cortex

Don't know?

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Term 🔊

Spinal Pathways:

What are the "ascending tracts?"

Excellent!

1. Corticospinal tracts
2. Corticobulbar tracts

1. Tectum
2. Tegmentum
3. Cerebral Aqueduct
4. Crus Cerebri (Basis Pedunculi)

1. Dorsal column-medial lemniscal pathway
2. Anterolateral system

1. Vestibulospinal tracts
2. Reticulospinal tracts
3. Rubrospinal tracts
4. Tectospinal tracts

Don't know?

Definition

1. Nucleus of inferior colliculus
2. Brachium of inferior colliculus
3. Cerebral aqueduct
4. Lateral lemniscus
5. Medial longitudinal fasciculus (MLF)
6. Trochlear nucleus
7. Central tegmental tract
8. Trigeminal and spinal lemnisci
9. Reticular formation of the tegmentum
10. Decussation of superior cerebellar peduncles
11. Medial lemniscus
12. Pontine nuclei and pontocerebellar fibers
13. Corticospinal, corticopontine, and corticobulbar fibers
(these collectively make up the Crus Cerebri or Cerebral Peduncle at this level)



You've got this!

Diagram: Caudal midbrain

Diagram: Rostral midbrain

Diagram: "Mesencephalon" Cross-Section at the Level of the Superior Colliculus

Midbrain Structures:

What does the Tectum consist of?

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Term 🔊

Midbrain Structures:

What does the Cerebral Aqueduct consist of?

You're doing great!

2 major ocular nerve tracts
which run directly adjacent

5 major gustatory nerve tracts which
intersect

3 major olfactory nerve tracts which
run parallel

4 major auditory nerve tracts which
converge

Don't know?

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Term 🔊

Midbrain Functions:

What are the midbrain's *associated senses*?

Nice work!

vision & hearing

hearing & smell

hearing & pressure

vision & smell

Don't know?

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Term 🔊

Midbrain Functions:The midbrain *is also called?*

Awesome!

Diencephalon

Metencephalon

Rhombencephalon

Mesencephalon

Don't know?

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Term 🔊

Midbrain Structures:

Where is the Tectum located?

lateral side of the midbrain

anterior side of the midbrain

ventral side of the midbrain

dorsal side of the midbrain

Don't know?

20 of 24

Term 🔊

Spinal Pathways:

What are the "descending pyramidal tracts?"

You've got this!

1. Vestibulospinal tracts
2. Reticulospinal tracts
3. Rubrospinal tracts
4. Tectospinal tracts

1. Dorsal column-medial lemniscal pathway
2. Anterolateral system

2 major ocular nerve tracts which run directly adjacent

1. Corticospinal tracts
2. Corticobulbar tracts

Don't know?

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Diagram: Rostral Midbrain

You're doing great!

1. Nucleus of inferior colliculus
2. Brachium of inferior colliculus
3. Cerebral aqueduct
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1. Superior Colliculus
2. Cerebral Aqueduct
3. Oculomotor Nucleus
4. Red Nucleus
5. Spinothalamic Tract and Medial Lemniscus (often grouped together in this view)

8. Interpeduncular Fossa

Don't know?

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Term 🔊

Diagram: Ascending and Descending Spinal Pathways

Nice work!

1. Superior Colliculus
2. Cerebral Aqueduct
3. Oculomotor Nucleus
4. Red Nucleus
5. Spinothalamic Tract and Medial Lemniscus (often grouped together in this view)
6. Substantia Nigra
7. Crus Cerebri (or Cerebral Peduncle)
8. Interpeduncular Fossa

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Don't know?

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Term 🔊

Can you differentiate the location of the following three structures: the midbrain, pons, & medulla

Awesome!

1. Superior Colliculus
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3. Oculomotor Nucleus
4. Red Nucleus
5. Spinothalamic Tract and Medial Lemniscus (often grouped together in this view)
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12. Red nucleus
13. Fibers of oculomotor nerve (CN III fibers)
14. Substantia nigra

Don't know?

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Term 🔊

Diencephalon Structures:

What does the Thalamus consist of (substructures)? (and its function?)

Excellent!

1. Red Nuclei
(Gait)
2. Medial Lemniscus
(somatosensation from the skin and joints)
3. Reticular Formation
(production of 5-HT, arousal, consciousness, motor coordination, & exhalation)

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